

SUBJECT	Lower Secondary Science Year 8
TOPIC	Motion and Pressure
ASSIGNMENT	Nine
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

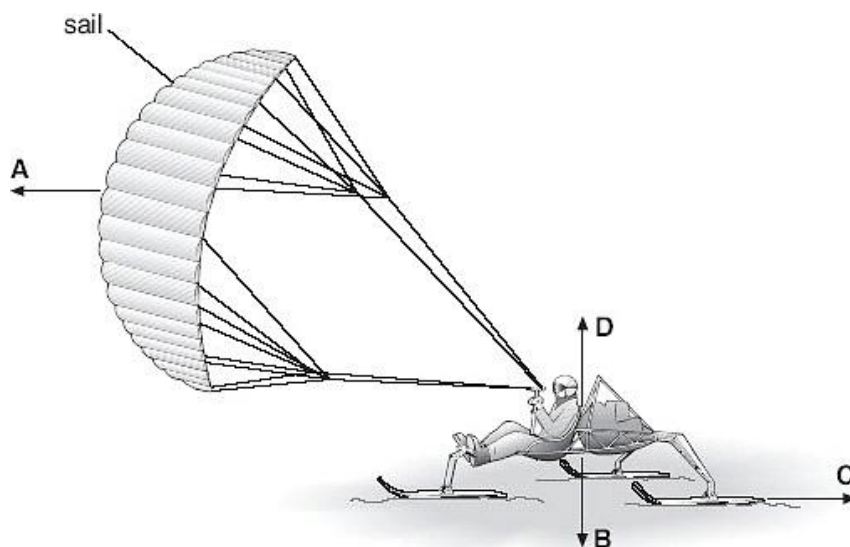
- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment Nine

MOTION AND PRESSURE

1. The drawing shows a snow-buggy being pulled by a sail.

The buggy rests on three skis on the snow.



- a. The drawing shows four forces that act when the snow-buggy is moving.

Draw a line from each force in the list below to the correct letter from the diagram.

Draw only three lines.

the weight of the buggy	A
the force pulling the buggy along	B
the friction between the skis and the snow	C
	D

(3 marks)

- b. A scientist travelled 80 kilometres (km) each day in the buggy.

How many kilometres did he travel in 10 days?

_____ km

(1 mark)

- c. The buggy carried the scientist, food and equipment for the journey.

The table shows how the total mass changed.

	total mass at start of journey (kg)	total mass at end of journey (kg)
mass of buggy, scientist, food and equipment	295	130

The buggy sank deeper into the snow at the start of the journey than at the end.

Why did it sink deeper at the start? Use the table to help you.

(1 mark)

- d. The buggy rests on three skis instead of three wheels.

Why are skis better than wheels for travelling on snow?

(1 mark)

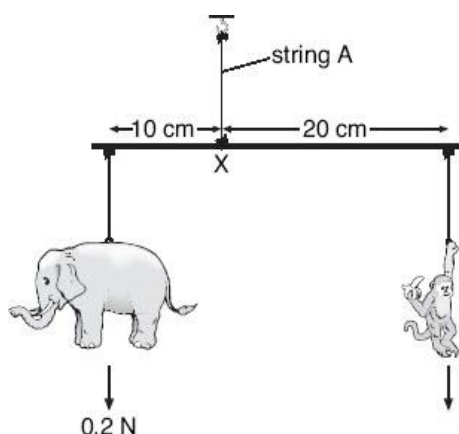
- e. When a bigger sail is used, the buggy goes faster.

How does a bigger sail help the buggy to go faster?

(1 mark)

(Total 7 marks)

2. A father makes a simple mobile for his young son. He uses plastic animals as shown below.



a.

i. The elephant weighs 0.2 N.

What is the turning moment produced by the elephant about point X?
Give the unit.

_____ (2 marks)

ii. What is the turning moment produced by the monkey about point X?

_____ (1 mark)

iii. What is the weight of the monkey?

_____ N (1 mark)

b. What is the size of the tension (force) in string A?

_____ N (1 mark)
(Total 5 marks)

3.

a. Megan was doing time-trials on her bike around a 400 metre horizontal track.

i. She took 32 seconds to travel 400 m.

What was her average speed? Give the unit.

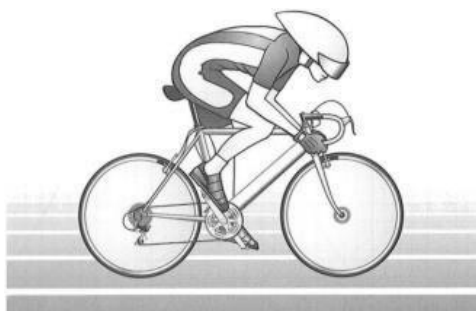
(2 marks)

ii. Compare the forward force on the bike with the backward force on the bike when Megan was travelling at a constant speed.

(1 mark)

b. Megan then crouched down over the handlebars to make herself more streamlined, as shown below.

She continued to pedal with the same force as before.



i. Compare the forward and backward forces on Megan and her bike now.

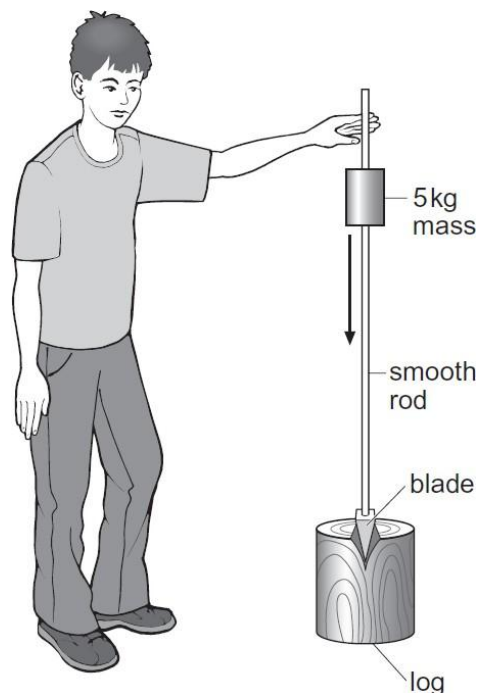
(1 mark)

ii. Explain your answer.

(1 mark)

(Total 5 marks)

4. David uses a falling mass to split wooden logs. The 5kg mass slides down the rod and hits the metal blade. The force on the blades splits the log.



- a. To lift the mass David uses energy stored in his muscles. What energy transfer occurs when David's muscles lift the mass?

From _____ energy in his muscles to gravitational potential energy of the mass.

(1 mark)

- b. David lifts the mass. The mass gains 50J of gravitational potential energy. The falling mass changes this energy into kinetic energy.

- i. As it falls, what is the maximum amount of energy the mass can change from gravitational potential energy to kinetic energy?

_____ J

(1 mark)

- ii. Not all the gravitational potential energy is transferred to kinetic energy as the mass falls. Give one reason for this.

(1 mark)

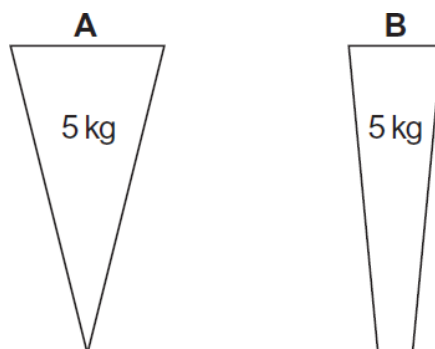
- c. Give **two** ways David can increase the kinetic energy of the mass just before it hits the blade.

1. _____

2. _____

(2 marks)

- d. David can use a different blade to split the logs. The diagram below shows two different blades, A and B.



The formula for pressure is:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

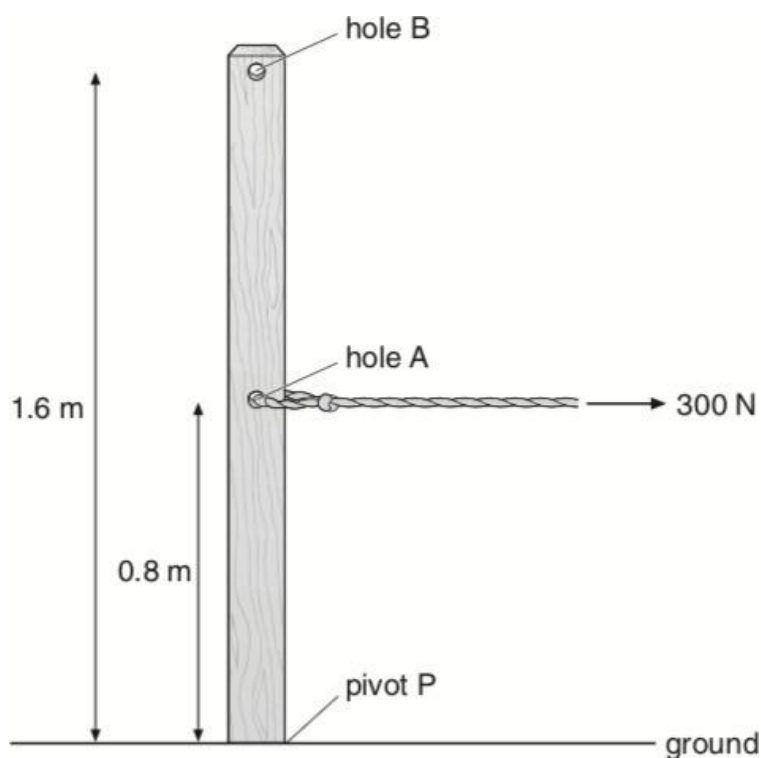
Which blade puts more pressure on the log? Write the letter.

Explain your answer in terms of area. Use the formula to help you.

(1 mark)

(Total 6 marks)

5. A builder tried to remove a wooden post from the ground by pulling with a rope.



a.

- i. The builder attached a rope to hole A, 0.8m above the ground. He pulled with a horizontal force of 300N.

Calculate the turning moment about the pivot P. Give the unit.

_____ (2 marks)

- ii. He then attached a rope to hole B, 1.6m above the ground. He pulled with a horizontal force. What force would produce the same turning moment as before?

_____ N (1 mark)

- b. The post breaks off and falls to the ground as shown.



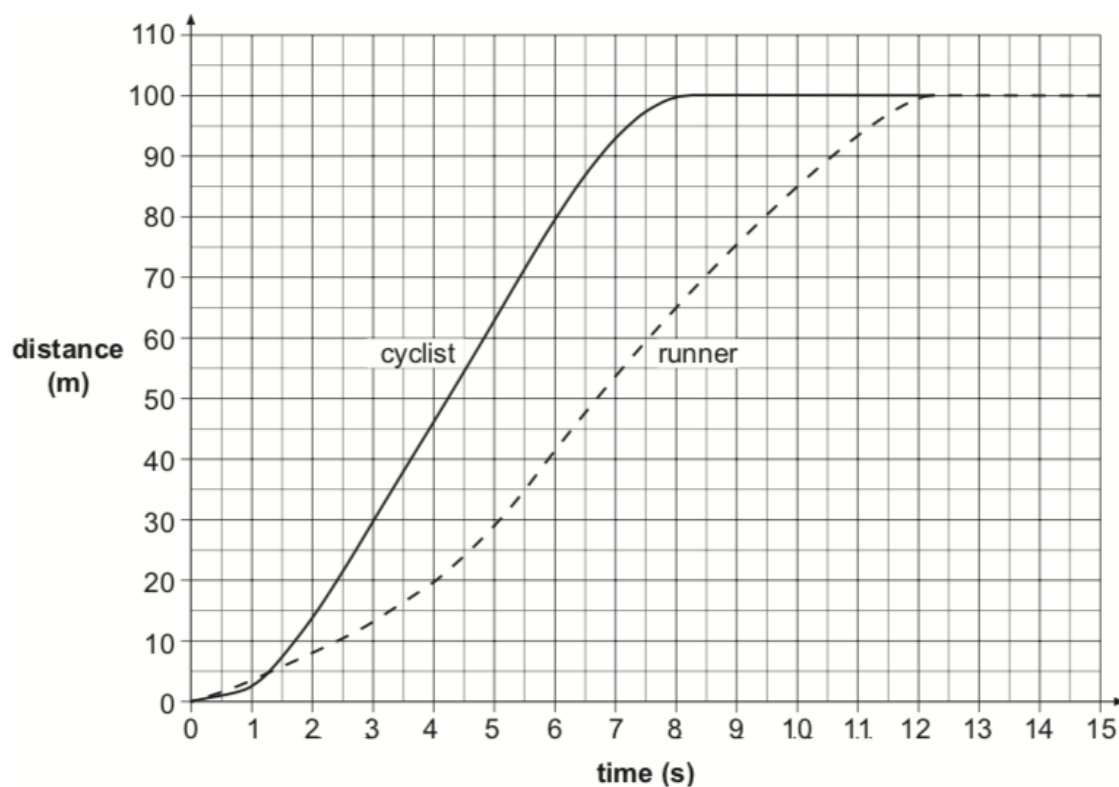
The weight of the broken post is 120N. The area in contact with the ground is 0.2m².

Calculate the pressure of the broken post on the ground. Give the unit.

(2 marks)

(Total 5 marks)

6. A cyclist and a runner have a race. The distance-time graph for the race is shown below.



- a. Use the graph to answer the following questions.

- i. How much time did it take the cyclist to travel 100m?

_____s

(1 mark)

- ii. When the cyclist finished the race how far behind was the runner?

_____m

(1 mark)

- iii. How much more time did the runner take compared with the cyclist to complete the race?

_____s

(1 mark)

- b. The cyclist is travelling at a constant speed between 3 seconds and 6 seconds.

How does the graph show this?

(1 mark)

c.

- i. When the race started, a walker set off at a steady speed of 2m/s. **Draw a line on the graph** to show the distance covered by the walker in the first 15 seconds. Use a ruler.

(1 mark)

- ii. Calculate how much time it will take for the walker to walk 100m.

_____s

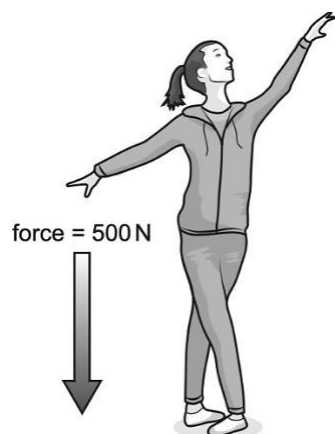
(1 mark)

(Total 6 marks)

7. Mia is a dancer.

The force she exerts on the floor is 500 N.

- a. She stands with both feet flat on the floor.



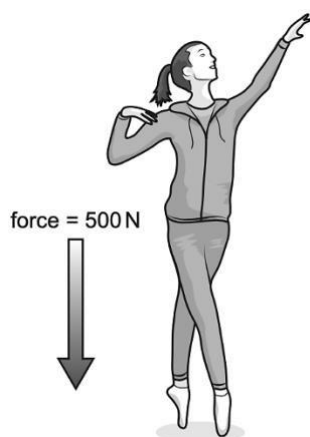
The area of her shoes in contact with the floor is 400 cm^2 . Calculate the pressure she exerts on the floor.

pressure..... N / cm^2

(2 marks)

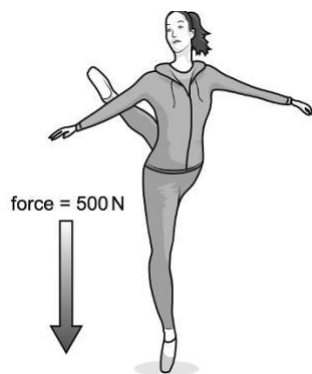
- b. Mia stands on the points of her feet.

Her new area of contact with the floor is 10 cm^2 rather than 400 cm^2 .



Describe what happens to the pressure she exerts on the floor.

(1 mark)



- c. Mia balances on one point.

She cannot remain on one point for a long time because it may damage her toes.

Suggest why this may damage her toes.

(1 mark)

(Total 4 marks)

TOTAL FOR ASSIGNMENT 38 MARKS